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Research Articles

Evaluating urban policy: A GIS-based comparison of urban green space suitability model with Adama City's Structural Plan

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ABSTRACT

Urban green spaces (UGS) are critical for mitigating the negative impacts of rapid urbanization, yet land use changes and inadequate planning persistently threaten them. This study addresses this dual challenge in Adama City, Ethiopia, by first developing a scientifically robust site suitability model and then using it to evaluate the City's existing urban policy. We quantified land use and land cover (LULC) dynamics from 1991 to 2021, revealing unsustainable urban sprawl as built-up areas expanded by 5,035.11 ha, a relative increase of 530.68%. This expansion occurred primarily at the expense of Farm land, and Shrub and bushes, which recorded absolute losses of 2,178.42 ha and 839.98 ha, respectively. These figures represent a relative decline of 9.17% for Farm land and a more significant 29.66% for Shrub and bushes. We then developed a GIS-based Multi-Criteria Analysis (MCA) model, integrated with the Analytical Hierarchy Process (AHP), to identify optimal sites for UGS development based on nine environmental, physical, and socio-economic criteria. The model classified 6.39% of the city as highly suitable and 25.41% as moderately suitable for UGS. Crucially, we compared these evidence-based results with the green space allocations in Adama's Proposed Structural Plan (2017) by considering Plan Implementation Evaluation (PIE) theory. The analysis revealed a severe misalignment: only 26% of planned green space lies in suitable zones, while 74% is located in less suitable to unsuitable areas. Furthermore, the plan allocates only 17% of the total Adama City's area to green space, falling short of the Ethiopian national 30% standard. This study highlights a critical gap between planning policy and scientific evidence in Adama City, demonstrating the urgent need to institutionalize geospatial suitability analysis within urban planning processes to foster sustainable and equitable urban development.

Introduction

The global trajectory of urban development is increasingly defined by the swift conversion of rural hinterlands into dense metropolitan hubs[1]. This phenomenon is exceptionally pronounced in emerging East African economies like Ethiopia, where rapid horizontal expansion often outpaces formal land-use planning. While offering socioeconomic opportunities, rapid and often unplanned urban growth, particularly in the Global South, leads to severe environmental degradation and pressure on essential services[2,3]. In Ethiopia, urban expansion is characterized by rapid horizontal growth and shifting spatial patterns, as evidenced by studies in Addis Ababa that show a dominance of edge expansion and fluctuating infill growth over the last three decades[4].

Ecological infrastructure, specifically urban green space (UGS), is frequently the primary victim of this accelerated expansion. These areas, comprised of natural or semi-natural vegetation, provide indispensable ecosystem services that are vital for urban resilience[5–7]. These spaces are vital for air and water filtration, microclimate regulation, biodiversity conservation, and public health, offering venues for recreation, social interaction, and mental well-being [8]. In the specific context of Adama, this loss is not merely an environmental concern but a threat to the city's ability to manage the microclimatic challenges of the Ethiopian Rift Valley.

Despite these critical local and global benefits, UGS in the region are continually degraded due to urban encroachment, lack of strategic site selection, and insufficient integration into urban planning frameworks [9,10]. This loss is not merely an environmental concern but a matter of spatial equity and environmental justice, as vulnerable populations often endure the most of diminished access to nature's benefits [11]. In rapidly urbanizing Ethiopian cities like Adama, studies have shown a

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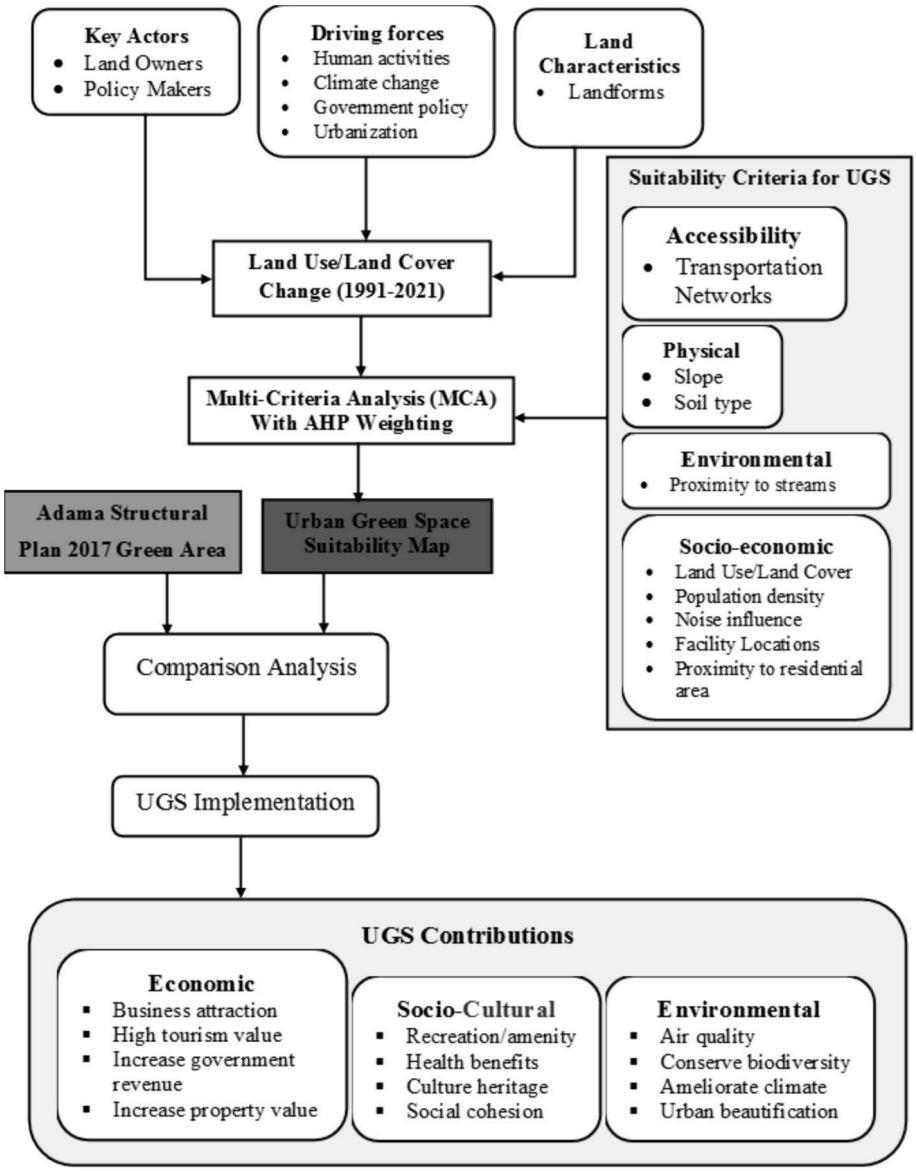


Fig. 1. Conceptual framework for the study.

disproportionate expansion of built-up areas compared to green spaces, creating an imbalance that threatens urban sustainability and resilience [12].

Strategic, evidence-based planning is essential to counteract this trend. Land Suitability Analysis (LSA) serves as a systematic technique for determining the geographic fitness of land for specific development goals[13]. When integrated with Geographic Information Systems (GIS) and Multi-Criteria Analysis (MCA), LSA becomes a powerful method for handling complex spatial decisions involving diverse environmental and socio-economic factors[14,15]. The Analytical Hierarchy Process (AHP) is a widely used MCA technique that allows for the systematic weighting of various criteria based on their relative importance[16,17]. While these methods are commonly applied in developed countries for UGS planning[18–20], their use in guiding urban policy in developing nations like Ethiopia remains limited.

Previous research in Ethiopia has often focused on documenting the loss of green spaces or evaluating existing UGS, with less emphasis on developing forward-looking, evidence-based siting tools that can directly inform urban policy[21] and comparing the mismatch between the proposed city plans with the results of the selected sites and national standards. This study aims to bridge this critical gap. We pose a fundamental question for urban policy evaluation: Does existing statutory planning align with scientific evidence for optimal land use? To bridge the gap between spatial modeling and policy action, this study adopts the Plan Implementation Evaluation (PIE) Theory [22,23]. This theoretical lens allows for a rigorous assessment of the conformity between statutory urban planning and scientific land suitability, identifying spatial discrepancies that threaten long-term urban resilience[24]. By moving beyond a simple inventory of green spaces to a formal evaluation of policy performance, this research addresses the critical implementation gap often found in rapidly urbanizing Global South contexts.

Urbanization in the Global South often outpaces the capacity of local governments to manage growth. While many cities create ambitious structural plans, a gap frequently exists between these paper policies and the ecological reality on the ground. This study uses Plan Implementation Evaluation (PIE) theory to analyze this gap. Researchers must look beyond the simple existence of a plan to see if that plan actually works. Most existing research in Ethiopia focuses only on current land use or the general benefits of green space which fail to quantify the specific spatial discrepancies between planning intentions and scientific suitability.

Evidence-based planning requires the application of scientific data to

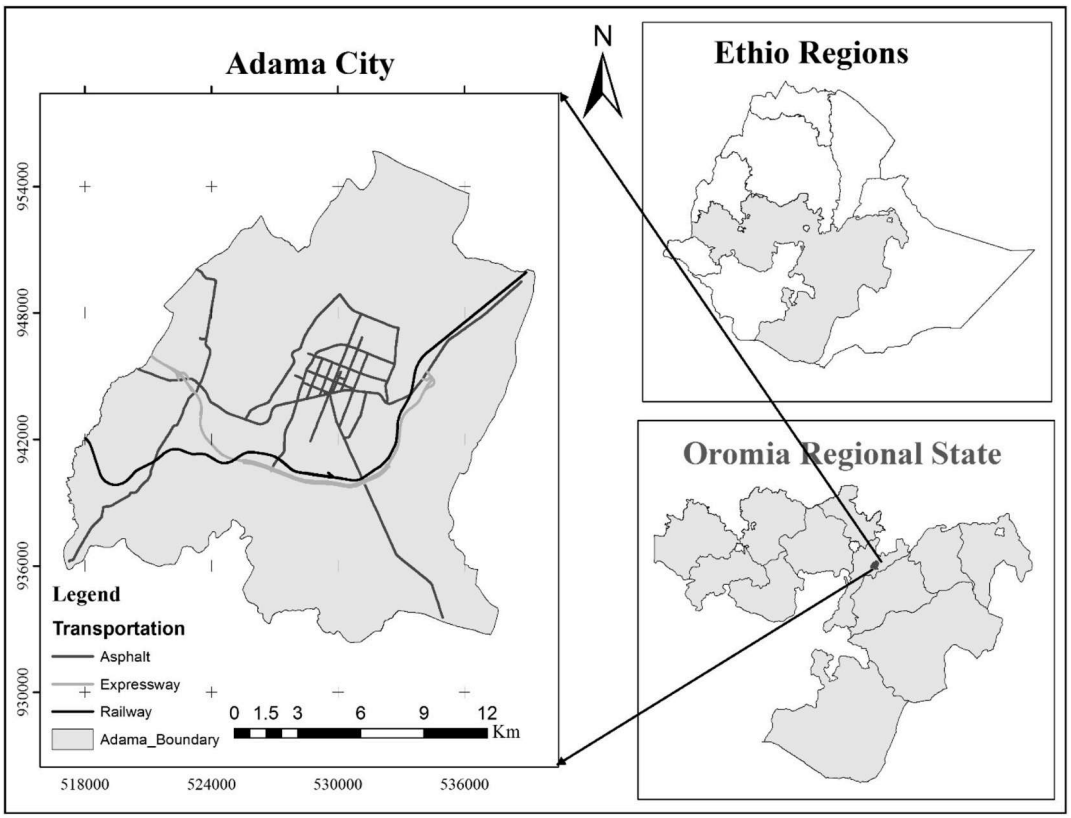


Fig. 2. Location Map of the Study Area.

verify official policies. When site suitability is ignored, plans are created that are inherently difficult to implement. This results in green spaces being placed on steep slopes or in floodplains where they provide limited recreational value and poor accessibility for citizens, despite their inherent ecological regulating services. By anchoring the study in policy evaluation, this study moves from showing basic description of Adama to a critical audit of its planning performance, using spatial data to reveal how the 2017 Adama City Structural Plan fails to align with environmental needs. This approach provides a replicable model for other rapidly growing cities to evaluate their own implementation failures and protect their ecological future.

To answer this, the study has three interconnected objectives: First, to quantify LULC changes (1991–2021) and establish the extent of ecological infrastructure loss particularly green space due to horizontal sprawl. Second, to develop a spatially explicit suitability model for UGS development using a GIS-based MCA-AHP framework, validated through expert panel consensus. Third, to perform a comparative analysis identifying convergences and critical misalignments within the Adama Proposed Structural Plan (2017), thereby calculating a Spatial Conformity Index to evaluate planning performance.

The research is anchored in Plan Implementation Evaluation (PIE) theory, integrating multi-temporal LULC analysis and GIS-based Analytical Hierarchy Process (AHP) to conduct a spatial audit of the 2017 Adama City Structural Plan. General conceptual framework for this study is presented on (Fig. 1).

Table 1
Summary of key data sources and software used.

Data/Software	Source/Version	Purpose
Landsat 5,7,8	USGS	LULC classification and change detection (1991, 2006, 2021).
ASTER DEM	Geospatial Information Inst.	Derivation of slope.
Road, Stream, Facilities	Google Earth Pro	Digitization of accessibility and proximity criteria.
Population Data	CSA	Creation of population density layer.
Soil Data	FAO	Soil suitability classification.
Structural Plan (2017)	Adama City Administration	Policy layer for comparative analysis.
ERDAS IMAGINE	2015	Satellite image preprocessing and classification.
ArcGIS	10.8	Spatial analysis, criteria mapping, weighted overlay, and comparison.
IDRISI Selva	32	AHP pairwise comparison and weight derivation.

Materials and methods

Study area

Adama City is one of the fastest growing cities in Ethiopia. The city is located between 8° 25′ 56″ N to 8° 38′ 49″N latitude and 39° 9′ 4″E to 39° 21′ 39″E longitude, situated in the Great Ethiopian Rift Valley. The City's topography is generally flat to gently sloping, with elevations ranging from 1,443 m to 2,055 m. The study area covers approximately 313.04 km2 within the City's proposed boundary (Fig. 2). Adama has experienced rapid population growth from 77,237 in 1984 to a projected 433,046 by 2011, driving significant and often unplanned urban expansion, which forms the core context for this investigation. Various advantages make Adama city appropriate selection for this study. Among these, the 2017 Structural Plan was designed to steer development through 2027; consequently, the 2021 LULC data utilized in this study serves as a strategic mid-term audit, allowing for an empirical evaluation of the plan's implementation progress and its alignment with ecological realities. In addition, it is a primary industrial hub and a transport corridor in the Oromia Regional State, Ethiopia, located approximately 99 km southeast of the capital city Addis Ababa. Its growth rate makes it a critical study area for studying the conflict

Table 2
Random Consistency Index (RI).

n	1	2	3	4	5	6	7	8	9	10
RI	0	0	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49

Source. Saaty [14].

between rapid infrastructure development and environmental preservation.

The Adama City Structural Plan evaluated in this study was officially ratified in 2017 with an effective period through 2027. The suitability model utilizes 2021 satellite imagery to serve as a strategic mid-term audit. While the plan was formulated using 2015–2016 baseline data, the 2021 LULC data allows for a fair evaluation of how the city has grown in the first four years of the plan's implementation cycle. This temporal alignment ensures that the identified mismatch reflects implementation gaps rather than just outdated base maps.

Data sources and software

The analysis utilized both spatial and non-spatial data from primary and secondary sources (Table 1). Key datasets included Landsat satellite imagery (1991, 2006, and 2021) from the USGS for LULC analysis. A 20 m resolution ASTER Digital Elevation Model (DEM) for deriving slope, road networks, streams, and locations of facilities (schools, health centers) and factories digitized from Google Earth Pro; population data from the Central Statistical Agency (CSA); soil type data from the Food and Agriculture Organization (FAO); and the digitized Adama Proposed Structural Plan (2017) obtained from the City administration. Analyses were performed using ERDAS IMAGINE 2015 for image processing, ArcGIS 10.8 for spatial analysis and modeling, and IDRISI Selva for AHP weight calculation.

Analytical framework

The methodology comprised three sequential phases: LULC change analysis, UGS suitability modeling, and policy comparison.

LULC change analysis (1991–2021)

To establish a historical baseline, Landsat images were preprocessed and classified using a supervised maximum likelihood algorithm into five classes: Built-up, Shrub and Bushes, Farmland, Bare Land, and Water Body. Classification accuracy was assessed using error matrices, with overall accuracies exceeding 93% and Kappa coefficients > 0.89 for all years, indicating strong agreement. Post-classification comparison was used to quantify changes between 1991, 2006, and 2021.

Gis-based multi-criteria suitability modeling

A GIS-MCA framework integrated with AHP was employed to identify optimal sites for UGS development. The process involved several steps:

1. Criteria Selection: As there are no universally agreed-upon criteria for UGS planning[25], we ensured the socio-economic and ecological relevance of the AHP weights by consulting a panel of 10 multidisciplinary experts. This panel consisted of four urban planners from the Adama City Planning Bureau, three environmental experts from Adama Science and Technology University, and three municipal administrators. This diverse stakeholder group allowed the model to balance ecological idealism with administrative feasibility. Furthermore, based on a synthesis of local expertise and a review of previous empirical studies [26–36], nine specific factors were selected to identify suitable sites for UGS development.

The reclassification thresholds for the nine criteria were derived from a synthesis of local planning standards provided by the Adama City Planning Bureau and empirical benchmarks from similar semi-arid climatic zones in the Global South [15,37]. For example, the Highly Suitable slope range (0–5%) aligns with national standards for cost-effective park construction and irrigation. These thresholds were further refined through a consensus-building session with the expert panel to ensure they reflect the local topographic and walkability constraints of Adama.

2. Standardization: All vector criteria were converted to raster format and reclassified onto a common suitability scale from 1 (Permanently Unsuitable – N2) to 5 (Highly Suitable – S1), following FAO land suitability guidelines[38]. The specific thresholds and buffer distances for each criterion were established based on local planning standards and refined through the expert panel's consensus.
3. **Weight Assignment using AHP:** The relative importance of each criterion was determined using AHP. A pairwise comparison matrix was constructed using Saaty's 1–9 scale [16] in IDRISI Selva 17. The computed weights were validated by a Consistency Ratio (CR) of 0.04, which is well below the acceptable threshold of 0.1 [39], confirming the reliability of the judgment. The consistency ratio for parameters pairwise comparison was determined as (Equation (1).

$$CR = \frac{CI}{RI} \tag{1}$$

Where CI represents consistency Index and RI represents random consistency Index.

Random consistency Index (RI) is the index whose value depends on the number (n) of factors being compared; as illustrated in (Table 2) [14]. Consistency index was the measure of how all targeted parameters were interrelated and calculated as (Equation (2).

$$CI = \frac{\lambda max - n}{n - 1} \tag{2}$$

Where λ max is the principal eigen value; n is the number of factors. λmax = Σ of the products between each element of the priority vector and column totals.

4. **Model Aggregation:** The standardized and weighted criterion maps were combined using a Weighted Linear Combination (WLC) in ArcGIS to produce a final composite suitability map with five classes: S1, S2, S3, N1, and N2.
5. **Weighted Overlay Analysis:** After the criteria maps and weights have been developed and established, a decision rule of multi-criteria analysis was used. There are three common decision rules in multi-criteria analysis namely weighted linear overlay, boolean overlay and ordered averaging [40,41]. In this study the weighted linear combination (WLC) technique was applied to aggregate the standardized layers. In this technique, factors or parameters (Xi) are multiplied by weight of the parameters (Wi) to get composited weights and then summed. WLC technique is a straightforward linear method calculating composite weights. WLC multiplies and sums up the layers to produce suitability maps for green space. This can be done by using the following formula to derive the intended map (i.e., urban green spaces suitability map for Adama City) and calculated as (Equation (3).

$$SI = \sum Wi{*}Xi \tag{3}$$

Where; SI is suitability index, Wi is weight of factor I, and Xi is normalized criterion of factors and $\sum$ is their summation.

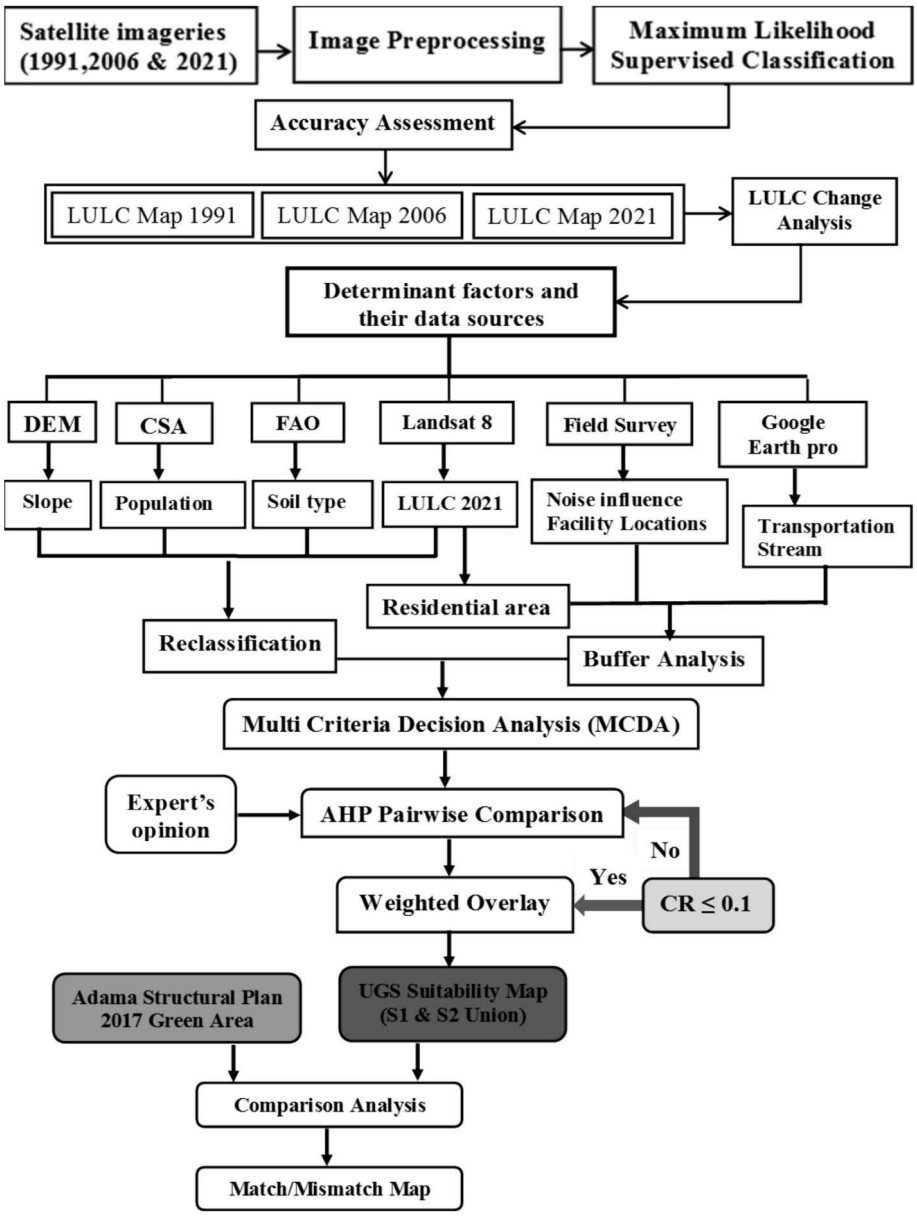


Fig. 3. Technological Scheme of the Study.

Sensitivity analysis

Sensitivity Analysis (SA) determines the reliability of the model through assessment of uncertainties in the reproduction results [42]. It is well-known that SA is crucial to the validation and calibration of numerical models. SA can be used as a tool to check the robustness of the final outcome against slight changes in the input data. Thus, SA can help reduce uncertainty in how a MCDM method operates and the stability of its outputs by illustrating the impact of introducing small changes to specific input parameters on evaluation outcomes [43].

In this study to validate the stability of the MCA output, a Sensitivity Analysis (SA) was carried out by changing the weights of the two most influential criteria LULC (from 29 to 39) and proximity to Built-up Area (from 21 to 11). By adjusting the weights of LULC and Residential area, the two highest-weighted factors, the model showed high spatial consistency. The negligible change in the Highly Suitable (S1) and Moderately Suitable(S2) class despite weight fluctuations indicates that the identified optimal sites are geographically robust and not merely a result of subjective weight assignment.

Model Match/Mismatch and policy analysis

The designated green area polygons (including parks, botanical gardens, urban agriculture, etc.) were extracted from the digitized 2017 Structural Plan. This policy layer was spatially overlaid with the final evidence-based suitability map using the Intersect tool in ArcGIS. The analysis quantified the area and percentage of planned UGS falling within each suitability class (S1 to N2). This provided a direct, measurable assessment of the alignment between policy-driven allocations and science-based suitability.

The technical procedure, involving data acquisition, AHP weighting, and spatial intersection, is detailed in the technological scheme (Fig. 3).

Results

LULC dynamics and urban expansion

The accuracy assessment yielded overall accuracies of 93.87% (1991), 96.56% (2006), and 94% (2021), with Kappa coefficients of 0.895, 0.952, and 0.921, respectively. These results confirm a strong

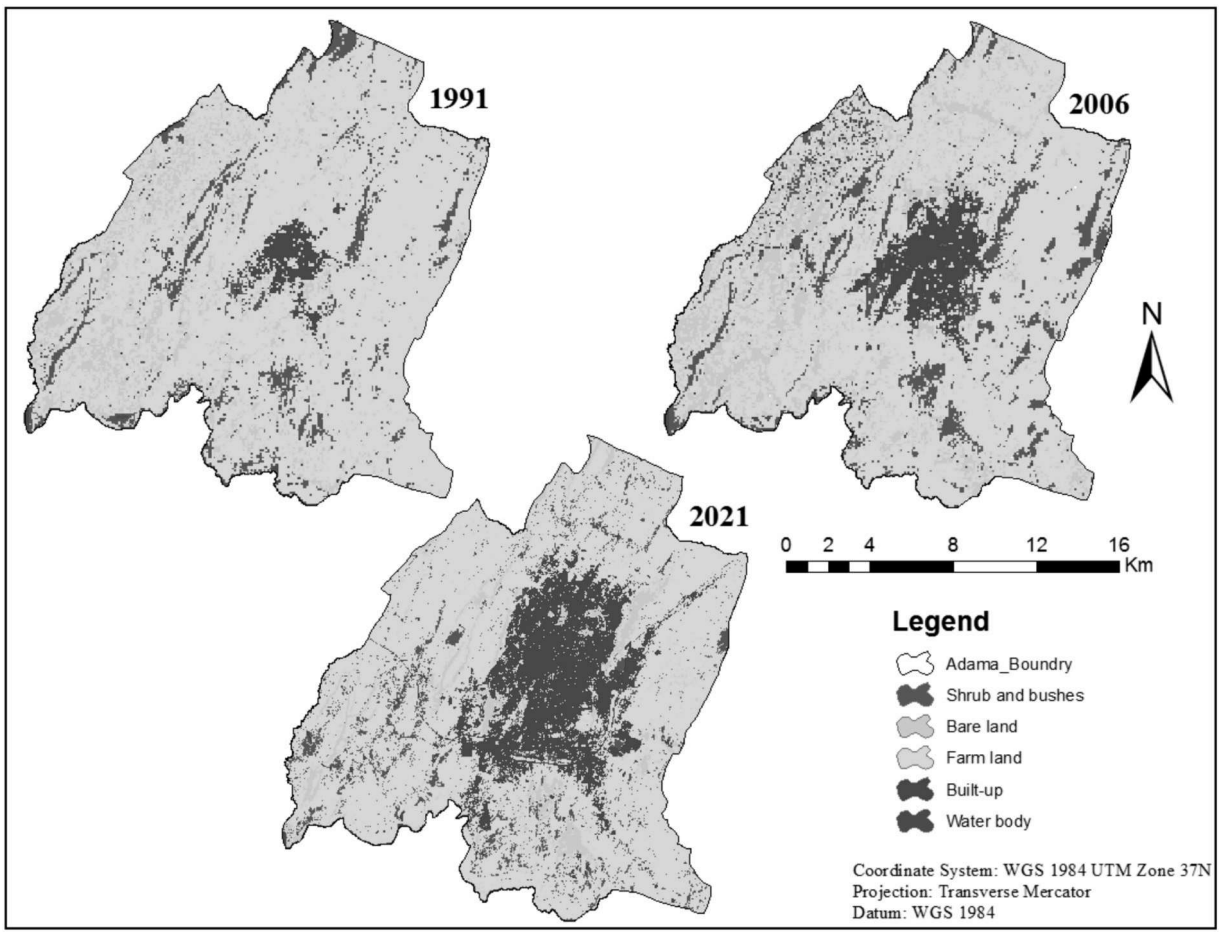


Fig. 4. LULC map of study area in 1991, 2006 and 2021.

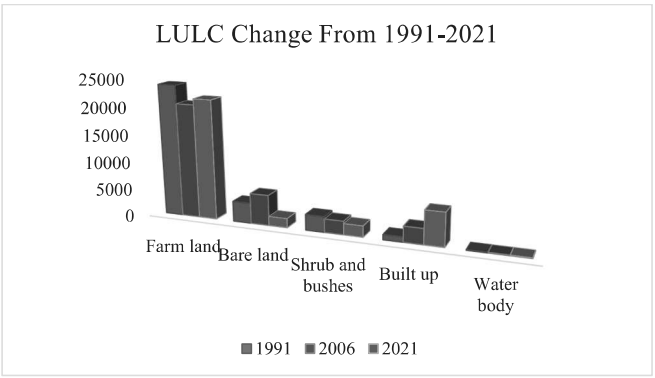


Fig. 5. Change Trends of LULC from 1991 to 2021.

agreement and high reliability of the produced LULC maps.

The LULC analysis revealed a dramatic transformation of the Adama landscape over the past 30 years (Fig. 4). The most significant change was the continuous increment of the built-up area, which grew from 948.8 ha (3.03%) in 1991 to 5,983.9 ha (19.12%) in 2021. This represents a total expansion of 5,035.11 ha, or a relative increase of 530.68%.

This urban growth occurred primarily at the expense of Farm land, and Shrub and bushes. Specifically, Farm land recorded the largest absolute loss of 2,178.42 ha, representing a relative decline of 9.17%. While Shrub and bushes saw a smaller absolute loss of 839.98 ha, they experienced a much more significant relative decline of 29.66% compared to their 1991 extent.

The result of LULC change analysis indicated that over the past 30 years built-up area showed continuous increment, whereas shrub and bushes showed continuous decline (Fig. 5). Consequently, those areas, which are close to built-up and urban centers, were subjected to significant loss of shrub and bushes (greeneries) in the last three decades. This trend of rapid urban expansion leading to green space depletion is a well-documented phenomenon, and these findings align with earlier studies that urbanization has a negative impact on green spaces [44,45].

This reduction of green spaces has indicated that the Adama City administration has a problem of proper planning and implementation to keep the city green with the exception of the establishment of a few roadside trees in the city. The results indicate that planning in Adama has prioritized construction over environmental conservation, underscoring the urgent need for a strategic approach to UGS development. Therefore, suitability analysis for urban green space development within the city are mandatory in order to maximize both quality and quantity of urban green spaces that bring value to urban dwellers and their environment in a sustainable way.

Site suitability for urban green space

Suitability of land use/land cover

Previous studies emphasize the importance of considering current land use/land cover (LULC) when selecting suitable sites for urban green space development, identifying varying levels of suitability based on land use type[15,20,26,28,31]). Drawing from literature analysis, bare land was deemed highly suitable, and shrub and bushes were considered moderately suitable for urban green space development. In this study, farmland was classified as less suitable, built-up areas as not suitable,

Table 3
Reclassified LULC and area coverage of suitability levels.

LULC type	Standardization Score/Value	Level of suitability	Area (ha.)	Area cover in (%)
Bare land	5	S1	1650.961	5.27
Shrub and bushes	4	S2	1992.234	6.36
Farm Land	3	S3	21568.945	68.90
Built-up area	2	N1	5983.909	19.12
Water body	1	N2	108.168	0.35

and water bodies as permanently unsuitable for green space development [15]. Converting bare land requires no removal of existing biomass and offers a blank canvas for planned green infrastructure. Shrublands in Adama represent existing ecological patches; assigning them a slightly lower suitability score (S2) reflects a conservation priority to preserve natural vegetation in its current state rather than clearing it for formal park development (Table 3; Fig. 6(A)).

Suitability of proximity to the stream

Earlier studies, such as those by[31], indicate that land closer to water bodies including rivers, lakes, reservoirs, and ponds is suitable for green space development. Such proximity is highly suitable from an ecosystem perspective, as these areas promote good environmental health. For this study, areas with a proximity to the stream less than 250 m are classified as highly suitable; those between 250 m and 500 m as moderately suitable; and those between 500 m and 750 m as less suitable (Table 4; Fig. 6(B)).

Suitability of proximity to transportations

Accessibility to urban green spaces is closely connected to their proximity to transportations. Road networks play a crucial role in

Table 4
Reclassified proximity to stream and area coverage of suitability levels.

Proximity tostream (m)	Standardization score/value	Level of suitability	Area (ha.)	Area cover in (%)
<250	5	S1	6398.858	20.44
250–500	4	S2	5451.442	17.41
500–750	3	S3	4429.110	14.15
750–1000	2	N1	3223.021	10.30
>1000	1	N2	11801.785	37.70

Table 5
Reclassified proximity to transportations and area coverage of suitability levels.

Proximity toRoad (m)	Standardization Score/Value	Level of suitability	Area (ha.)	Area cover in (%)
<500	5	S1	10386.607	33.18
500–1000	4	S2	5212.222	16.65
1000–1500	3	S3	3703.241	11.83
1500–2000	2	N1	3099.267	9.90
>2000	1	N2	8902.879	28.44

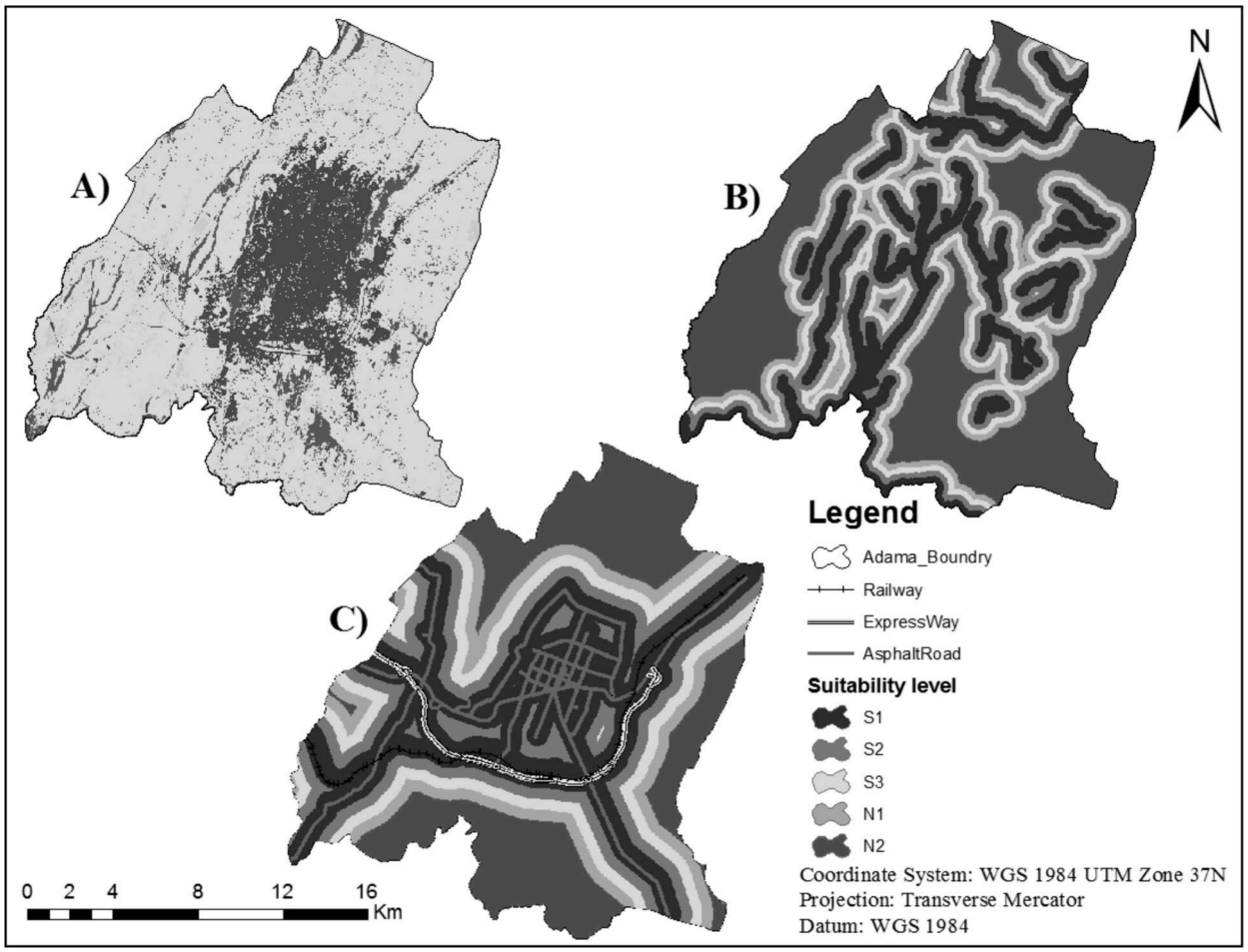


Fig. 6. Existing LULC (A), proximity to stream(B) and proximity to transportations(C) map.

Table 6
Reclassified slope and area coverage of suitability levels.

Slope (%)	Standardization Score/ Value	Level of suitability	Area (ha.)	Area cover in (%)
<5	5	S1	9745.398	31.13
5–10	4	S2	9549.294	30.51
10–15	3	S3	5128.518	16.38
15–30	2	N1	5051.256	16.14
>30	1	N2	1829.748	5.84

providing convenient and feasible access for the local population to reach nearby green areas[46,47]. Therefore, green space sites situated at an acceptable proximity to roads for ease of transport access are highly preferable[27,29]. In this study, using buffer analysis, areas within a 500 m radius of the road network were considered highly suitable; areas between 500 m and 1000 m were moderately suitable; and areas between 1000 m and 1500 m were considered less suitable (Table 5; Fig. 6 (C)).

Suitability of slope

Earlier studies have shown that low-slope areas are highly suitable for the development of urban green spaces[26,28,33,48]. The aim of building a green space is to create a common space for use by all sectors of society such as children, the elderly and the disabled; so, taking the necessary measures in this respect is essential[33]. Based on FAO [49], the slope value 0 to 0.2% is considered as flat and up to 5% gently slope and these slopes are considered as highly suitable in this study. Slopes between 5–10% and 10–15% are classified as moderately suitable and less suitable, respectively (Table 6,Fig. 7(D)).

Suitability of soil type

The soils of Adama City are classified into three principal types: Eutric Cambisols, Haplic Xerosols, and Lithosols (Table 7; Fig. 7(E)). Eutric Cambisols possess relatively good structure and chemical properties, making them less susceptible to significant degradation. Due to their increasing clay content with depth, Eutric Cambisols are considered highly suitable soils, as they are generally not greatly impacted by degradation. Haplic Xerosols, characteristic desert soils with low organic matter, are prone to wind erosion and the concentration of soluble salts; however, they are considered moderately suitable compared to the other types. Conversely, due to their lack of moisture and shallow profile, which preclude cultivation, Lithosols were considered unsuitable in this study compared to the other soil types [50].

Suitability of proximity to noise influence

Noisy areas, such as factory areas, are generally unsuitable for green space development due to high noise and smoke pollution[32,48]. Consequently, sites located farther from factories are given higher preference for green space development. In this study, by making buffer

Table 7
Soil type and area coverage of suitability levels.

Soil type	Standardization Score/Value	Level of suitability	Area (ha.)	Area cover in (%)
Eutric Cambisols	5	S1	17199.705	54.94
Haplic Xerosols	4	S2	3988.176	12.74
Lithosols	2	N1	10116.333	32.32

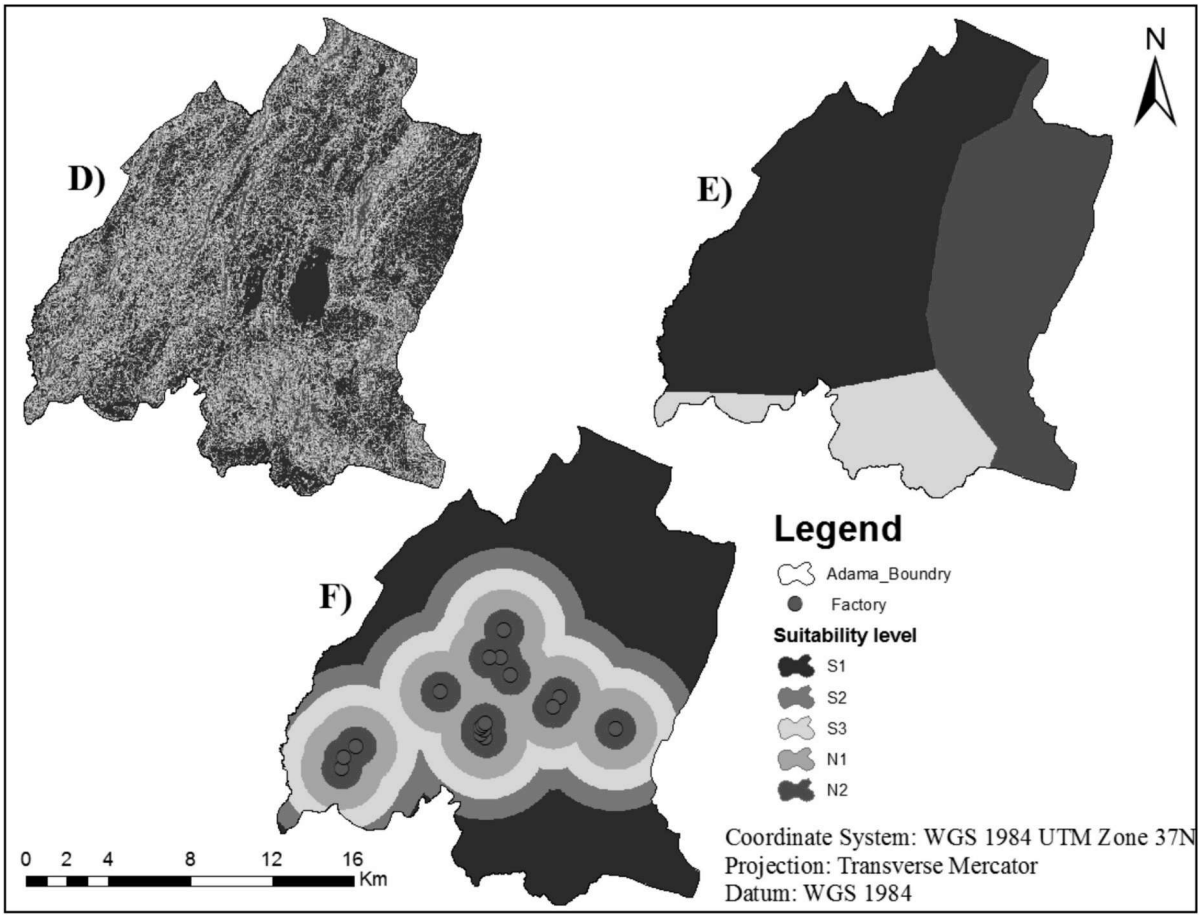


Fig. 7. Slope (D), Soil (E) and proximity to factory (F) map.

Table 8
Reclassified proximity to factory and area coverage of suitability levels.

Proximity tofactory (m)	Standardization Score/Value	Level of suitability	Area (ha.)	Area cover in (%)
>4000	5	S1	13606.094	43.46
3000–4000	4	S2	4114.139	13.14
2000–3000	3	S3	5180.503	16.55
1000–2000	2	N1	5429.615	17.35
<1000	1	N2	2973.863	9.50

Table 9
Reclassified population density and area coverage of suitability levels.

Population density (pop/ha)	Standardization Score/Value	Level of suitability	Area (ha.)	Area cover in (%)
>80	5	S1	1644.247	5.25
50–80	4	S2	820.223	2.62
20–50	3	S3	2587.729	8.27
5–20	2	N1	2781.405	8.89
<5	1	N2	23470.645	74.97

analysis, as totally unsuitable if less than 1000 m, not suitable between 1000 m and 2000 m, less suitable from 2000 m to 3000 m, moderately suitable from 3000 to 4000 m and highly suitable greater than 4000 m for green space development (Table 8,Fig. 7(F)).

Population density

Population density is a substantial socio-economic factor influencing the suitable selection of green space in urban areas. Several researchers [20,27–29,31] recommend that areas with high population density are highly suitable for developing green spaces. Consistent with this, the central parts of Adama City, being densely populated, are considered highly suitable for urban green space development. Conversely, the outer portion of the city is sparsely populated and is believed to be insufficiently suited for UGS development (Table 9,Fig. 8(G)).

Suitability of proximity to residential area

Different previous studies carried out by[15,30,51,52], recommend that green spaces must be accessible to settlement area in urban areas, since they have numerous ecological, social and economic benefits. Distance between settlement areas and green spaces had an adverse impact on users and reported that green spaces such as parks, playground and sport field closest to settlement areas are most popular. Thus, the proximity of green spaces to the settlement area in terms of distance is important to consider. For this study, areas within 500 m, 500 m to 1000 m and 1000 m to 2000 m, distances from the settlement area have been considered as highly suitable, moderately suitable and less suitable respectively (Table 10,Fig. 8(H)).

Suitability of proximity to public facilities

The facilities location covers the schools, college, major health facilities and hospital's locations that are crucial and important to get all services and meantime need some green spaces nearby such locations. Based on those views the lands which are closer to those facilities are more suitable than farther land[20]. Therefore, if proximity to the facilities location is less than 400 m, it was considered as highly suitable; 400 m – 800 m, moderately suitable; 800 m-1200 m, less suitable; 1200 m–1600 m, not suitable and > 1600 m, permanently not suitable for

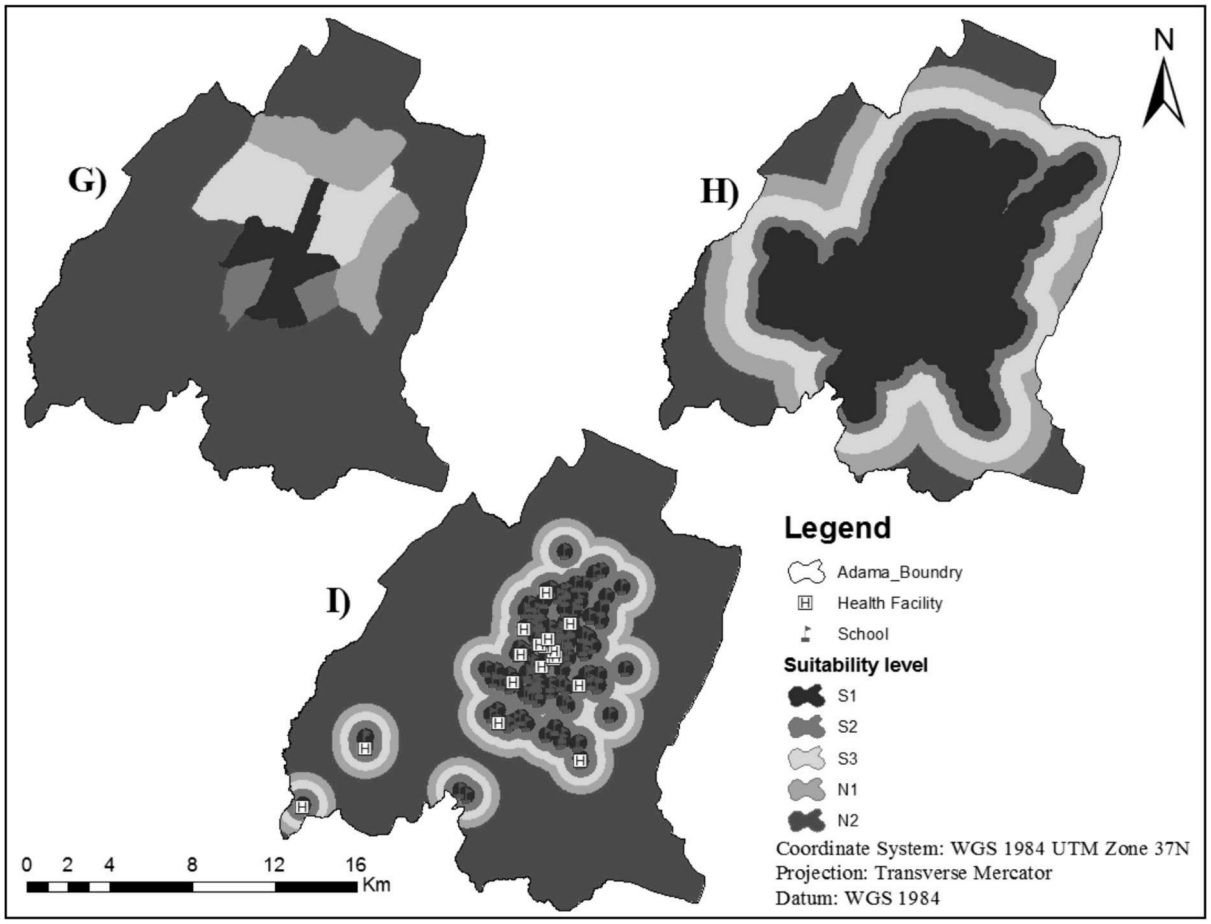


Fig. 8. Population density(G), residential area(H) and public facilities (I) map.

Table 10
Reclassified proximity to residential area and area coverage of suitability levels.

Proximity to residential area (m)	Standardization Score/Value	Level of suitability	Area (ha.)	Area cover in (%)
<500	5	S1	12609.880	40.28
500–1000	4	S2	3304.301	10.55
1000–2000	3	S3	5896.489	18.84
2000–3000	2	N1	4397.972	14.05
>3000	1	N2	5095.565	16.28

Table 11
Reclassified proximity to public facilities and area coverage of suitability levels.

Proximity to facilities location (m)	Standardization Score/Value	Level of suitability	Area (ha.)	Area cover in (%)
<400	5	S1	3366.321	10.75
400–800	4	S2	2913.730	9.31
800–1200	3	S3	2417.016	7.72
1200–1600	2	N1	2449.925	7.83
>1600	1	N2	20157.214	64.39

developing green space (Table 11; Fig. 8(I)).

Evidence-Based suitability for UGS development

The AHP analysis assigned the highest weights to LULC (29%) and proximity to residential area (21%)(Table 12). The consistency ratio was 0.04, confirming the reliability of the weights (Fig. 9).

Weighted overlay analysis

The final suitability map, created by overlaying the weighted criteria, categorizes the study area into five levels of suitability for UGS development (Fig. 10).

The results show that 2000.8 ha (6.39%) of the area is highly suitable (S1) and 7953.8 ha (25.41%) is moderately suitable (S2). A large portion, 11402.3 ha (36.42%), was less suitable (S3), with the remaining area being not suitable (N1) or permanently unsuitable (N2) (Table 13).

The highly and moderately suitable areas are predominantly located in current bare land and shrubland areas. Field verification confirmed that areas identified as suitable correspond with appropriate on-the-ground conditions for greening initiatives.

Critical Evaluation of the Structural Plan (2017)

The comparison between the suitability model and the statutory plan revealed significant shortcomings in current urban policy.

Table 12
Pairwise comparison matrix.

Criteria	LULC	RE	TR	NI	ST	SO	SL	FL	PD	Weight
LULC	1	2	3	3	4	4	6	6	7	29
RES	1/2	1	2	3	3	4	4	6	6	21
TR	1/3	1/2	1	2	3	3	4	4	6	15
NI	1/3	1/3	1/2	1	2	3	3	4	4	11
ST	1/4	1/3	1/3	1/2	1	2	3	3	4	8
SO	1/4	1/4	1/3	1/3	1/2	1	2	3	3	6
SL	1/6	1/4	1/4	1/3	1/3	1/2	1	2	3	4
PF	1/6	1/6	1/4	1/4	1/3	1/3	1/2	1	2	3
PD	1/7	1/6	1/6	1/4	1/4	1/3	1/3	1/2	1	3

Note: LULC = Land use/ land cover, RES = Residential area, TR = Transportations, NI = Noise Influence, ST = Stream, SO = Soil type, SL = Slope, PF = Public Facilities, PD = population

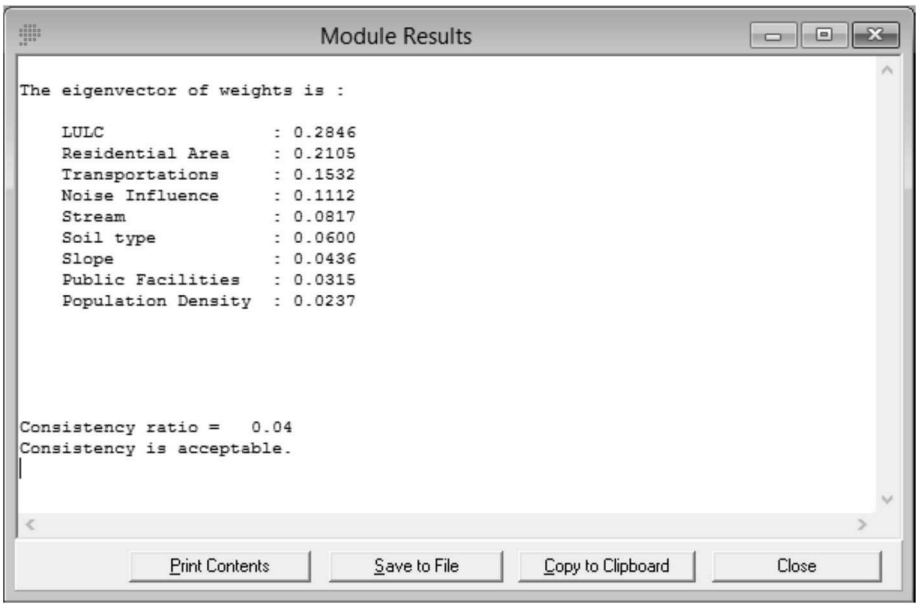


Fig. 9. Eigenvector of the pair wise comparison matrix.

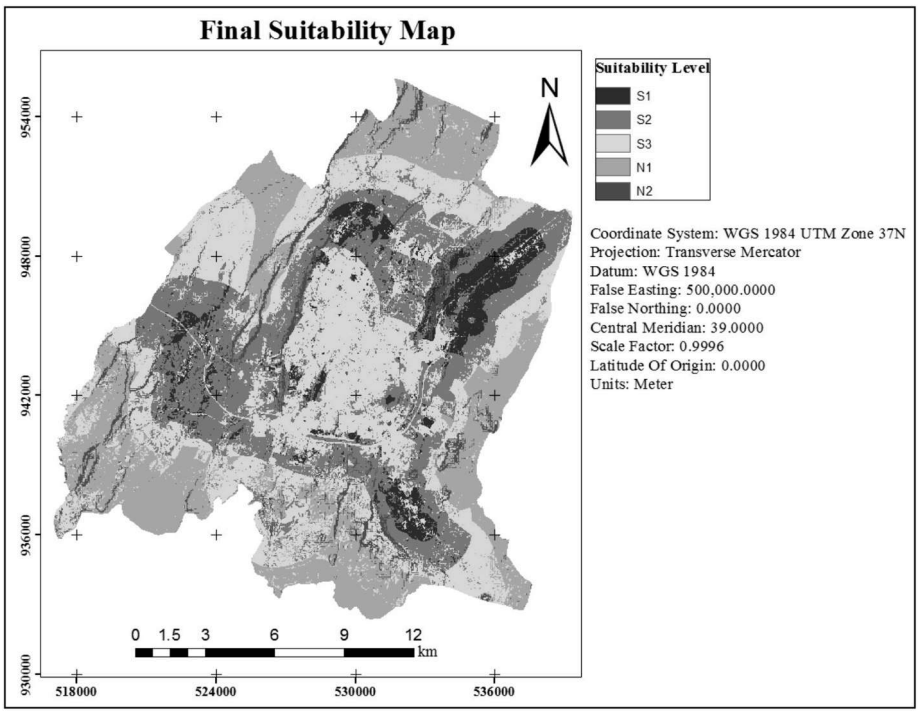


Fig. 10. Final suitability map.

Table 13
Final suitability area and percent of total area coverage.

Level of Suitability	Area (ha.)	Percent of total area (%)
S1	2000.805	6.39
S2	7953.809	25.41
S3	11402.344	36.42
N1	8159.420	26.07
N2	1787.837	5.71

1. **Quantitative Shortfall:** The 2017 Structural Plan allocates only 5,483.9 ha (17.52% of the City's area) for green space uses. This allocation falls substantially short of the 30% national standard set by the Ethiopian Ministry of Urban Development and Housing [53], indicating a policy that undervalues green infrastructure from the outset (Fig. 11).
2. **Severe Locational Misalignment:** The spatial overlay exposed a profound disconnect between planned and scientifically suitable locations (Table 14; Fig. 12L). A mere 25.92% of the planned green space area lies within suitable (S1 + S2) zones. The vast majority 74.08% is situated in areas classified as less suitable to unsuitable (S3 + N1 + N2) (Fig. 13).

This misalignment suggests that the plan's green space allocations were likely driven by factors such as leftover land, existing land tenure, or infrastructure corridors, rather than a systematic analysis of biophysical and socio-economic suitability.

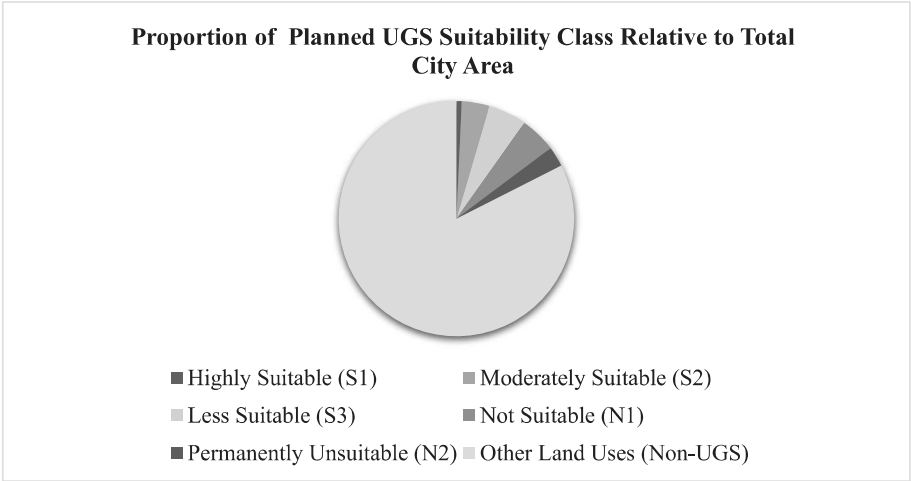


Fig. 11. Proportion of Planned UGS Suitability Class Relative to Total City Area.

Table 14
Spatial alignment of the 2017 Structural Plan's green areas with the suitability model.

Suitability Class	Area of Planned UGS (ha)	Percent of Total Planned UGS	Percent of Total City Area
Highly Suitable (S1)	230.9	4.24%	0.74%
Moderately Suitable (S2)	1,189.1	21.68%	3.80%
Total Suitable (S1 + S2)	1,420.0	25.92%	4.54%
Less Suitable (S3)	1,653.9	30.15%	5.28%
Not Suitable (N1)	1,555.3	28.34%	4.97%
Permanently Unsuitable (N2)	854.7	15.59%	2.73%
Total Unsuitable (S3 + N1 + N2)	4,063.9	74.08%	12.98%

Discussion

Spatial dynamics of green space loss and the institutional drivers of mismatch

The LULC trends confirm a global pattern observed in many secondary cities across the Global South, where economic and infrastructural development overwhelmingly trumps environmental conservation [31,54]. However, the severe misalignment identified in this study, where 74% of planned green space sits on unsuitable land, suggests that the problem in Adama is not merely a lack of GIS technology, but is rooted in deeper institutional and socio-economic drivers.

First, Land Tenure and Compensation Costs play a decisive role. Highly Suitable lands (S1) are often located on flat, accessible areas that are already under intense pressure from high-value real estate development. For municipal planners, designating these high-value areas as Green Space would require massive financial compensation to private landholders[37]. Consequently, UGS is often relegated to marginal lands, such as steep slopes or peripheral zones (N1/N2), where land value is lower and administrative resistance is minimal, regardless of ecological suitability.

Second, the Data Silo Phenomenon within the Adama City Planning Bureau likely contributed to this disconnect. The 2017 Structural Plan was likely developed using a cadastral-first approach, focusing on road networks and plot boundaries, rather than a landscape-first approach that integrates ecological layers [55]. This creates Paper Parks, zones that satisfy a quantitative regulatory quota on a map but are physically or financially impossible to implement on the ground.

Finally, the Construction-First Bias in Ethiopian urban policy prioritizes rapid housing and industrial hubs to accommodate the 530% built-up growth[56]. In this high-pressure environment, green space is often treated as leftover space, filling whatever gaps remain after industrial and residential zones are mapped, rather than being used as a foundational layer for urban resilience.

Results and policy Implications

The analysis reveals two critical findings for urban policy: a quantitative shortfall and a severe locational misalignment between the scientific model and the statutory plan. The core finding of policy-science misalignment is a significant contribution to urban planning discourse in Ethiopia. Placing green spaces in unsuitable locations (e.g., steep slopes, poor soils, floodplains, or areas far from residential populations) undermines their intended ecological functions, reduces their accessibility

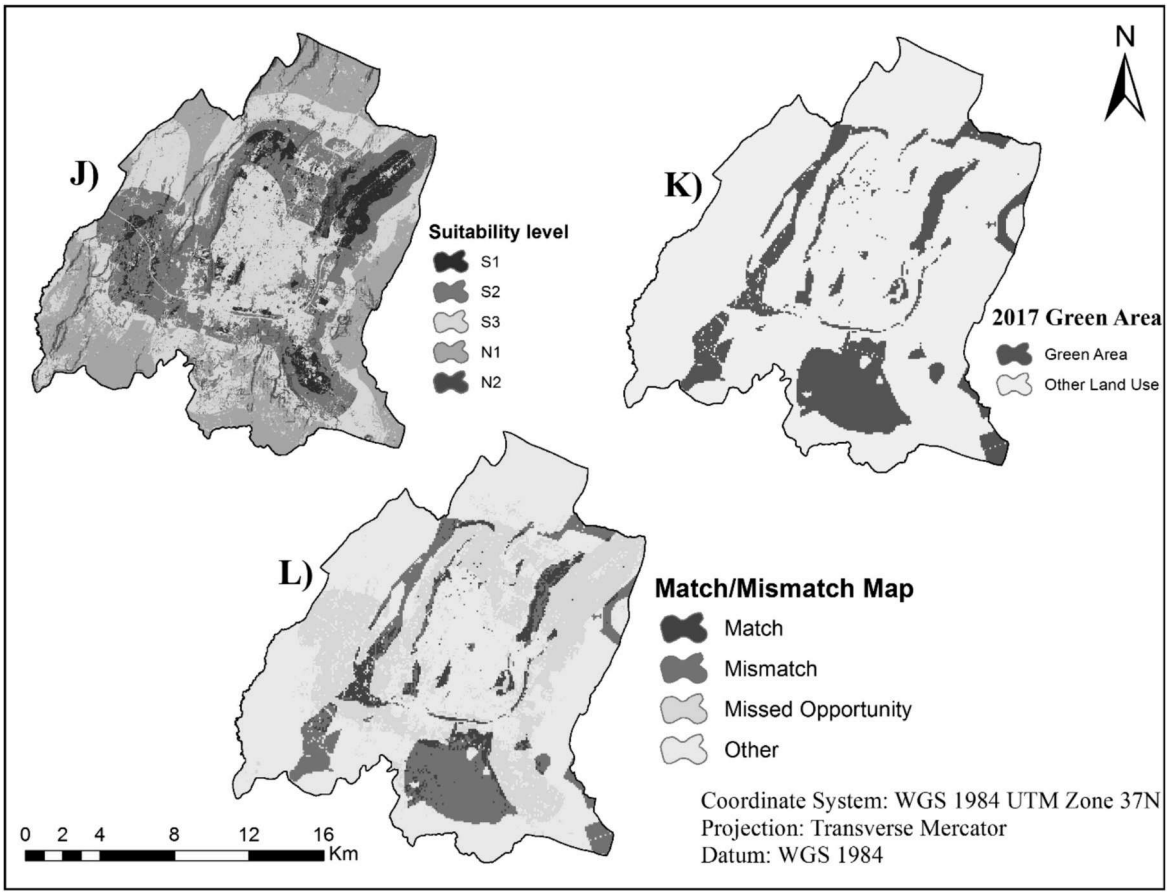


Fig. 12. Final suitability(J), Proposed 2017 green area(K) and Match/Mismatch (L) map. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

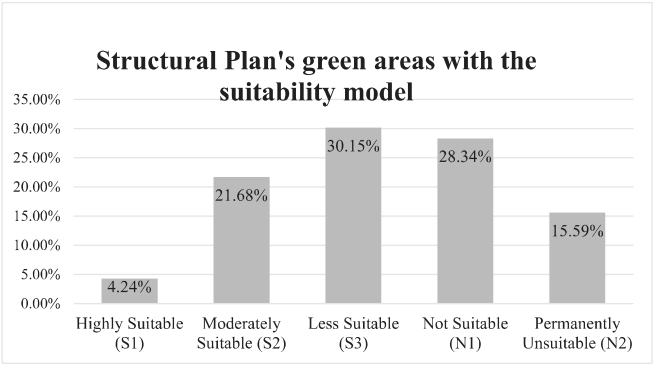


Fig. 13. Structural Plan's green areas with the suitability model. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 15.
Summary of Plan Evaluation Metrics.

Evaluation Metric	Statistical Value	Analytical Interpretation and Policy Impact
Spatial Conformity Index	26% Match	**High Policy Misalignment:** Only a quarter of the proposed green spaces align with scientifically optimal locations. This indicates that the official plan likely prioritized administrative convenience or leftover land over ecological and physical suitability.
Regulatory Compliance Gap	17.5% vs. 30%	**Target Shortfall:** The Structural Plan fails to meet the Ethiopian National Urban Greenery Standard of 30%. This quantitative shortfall creates a Green Equity Gap, limiting the city's long-term climate resilience and public health capacity.
Suitability Mismatch	74.16% (S3 + N1 + N2)	**Implementation Risk:** Nearly three-quarters of the planned UGS is situated on land classified as Less to Permanently Unsuitable. This suggests that if the plan is executed as written, these green spaces will face diminished ecological viability, prohibitive maintenance costs, or insurmountable physical construction barriers.

and usability, increases long-term maintenance costs, and ultimately leads to underutilization or failure[57,58]. This represents a lost opportunity to leverage green infrastructure for enhancing urban livability, social equity, and climate resilience.

The quantitative shortfall (17% vs. 30% standard) reflects a systemic undervaluation of green infrastructure in the face of development pressure. This is a common policy failure that requires not just technical correction but a shift in political and planning priorities to recognize UGS as essential public infrastructure, not optional amenities.

The low weight assigned to population density (3%) in our AHP model, while emphasizing land convertibility (LULC), reveals an inherent tension in UGS planning the trade-off between ecological restoration potential and immediate social equity. Prioritizing bare land and shrub areas is ecologically logical for creating large, connected green corridors[44]. However, this approach may not address the acute green space deficits in densely populated, built-up neighborhoods. Future planning should adopt a dual strategy: securing ecologically optimal sites for Citywide environmental benefits while also mandating smaller, interstitial green spaces in high-density zones to ensure equitable access for all residents [26,45].

Our findings resonate with studies in similar contexts. Research in Addis Ababa by Abebe and Megento [26]and other Ethiopian towns by Girma et al. [59] have also identified gaps between planned and optimal UGS, underscoring a widespread need for institutionalizing geospatial tools. This study provides a replicable methodology for other cities to audit their plans and transition towards evidence-based, sustainable urban development.

Consequently, the Proposed Structural Plan not only falls short in quantity (17.5% vs. the 30% national mandate) but also in spatial quality.

As shown in the (Table 15), the Spatial Conformity Index of 26% reveals a systemic disconnect. It suggests that urban planners in Adama may be practicing Top-Down mapping filling in the gaps left by residential and industrial zones with green space, rather than identifying the best land for ecology first. This results in an Implementation Risk where 74% of the planned parks are on unsuitable land (S3/N1/N2), such as steep slopes or poor soil. This creates the danger of Paper Parks areas that exist on official maps but are unlikely to survive or provide social value in reality [60,61].

Ecological fragility and flood risk

While proximity to water bodies is a positive suitability factor for recreational UGS, it introduces complexities regarding seasonal flood risks in the Rift Valley context. In this model, suitability primarily implies conservation and low-impact development suitability. We recommend that high-suitability zones near riparian buffers be managed as Blue-Green Infrastructure to mitigate flood risks rather than sites for heavy park infrastructure. This approach balances the social demand for water-proximate recreation with the need for ecological resilience.

Conclusion and recommendations

This study successfully developed a robust, evidence-based suitability model for urban green space in Adama City and used it as a critical lens to evaluate the City's prevailing structural plan. The analysis reveals a concerning disconnect: the plan not only allocates insufficient land for green space but also places the majority of it in suboptimal locations, thereby limiting its potential ecological and social benefits. This misalignment points to a broader gap between urban policy formulation and scientific spatial analysis.

To bridge this gap and steer Adama toward a more sustainable and equitable future, we recommend the following actionable steps for urban planners and policymakers:

1. **Institutionalize Geospatial Suitability Analysis:** The Adama City Administration and Urban Planning Bureau should formally adopt GIS-based MCA as a mandatory, standard step in the urban planning cycle, particularly for critical land uses like green infrastructure.
2. **Revise the Current Structural Plan:** The findings of this study provide a direct scientific basis for revising the 2017 plan. Priority should be given to reallocating and expanding green spaces into the identified Highly Suitable and Moderately Suitable zones, especially those near residential areas.
3. **Align with and Enforce National Standards:** The City must develop a clear strategy to increase its green space allocation toward the national standard of 30%, treating it as a non-negotiable component of urban resilience and public health.
4. **Build Technical and Planning Capacity:** Investment is needed in training planners and decision-makers in geospatial technologies, MCDA methods, and evidence-based planning principles.
5. **Integrate Equity into Suitability Modeling:** Future iterations of the suitability model should more explicitly incorporate social equity indicators (e.g., current access deficits, population density) to ensure green space development promotes justice alongside ecological goals.

By embracing evidence-based planning, Adama can transform its green spaces from afterthoughts in the development process into strategically located, functional assets that enhance ecological integrity, social well-being, and long-term urban sustainability. This approach offers a valuable model for other rapidly urbanizing cities facing similar challenges.

CRediT authorship contribution statement

Yirgalem Gerbaba Gurmessa: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Getinet Teshome Wendimagegn:** Writing – review & editing, Software, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics approval and consent to participate.

Not Applicable.

Consent for publication.

We have agreed to submit and approved the manuscript for submission.

Availability of data and material.

The datasets used and/or analysed are available from the corresponding author on reasonable request.

Data availability

Data will be made available on request.

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